

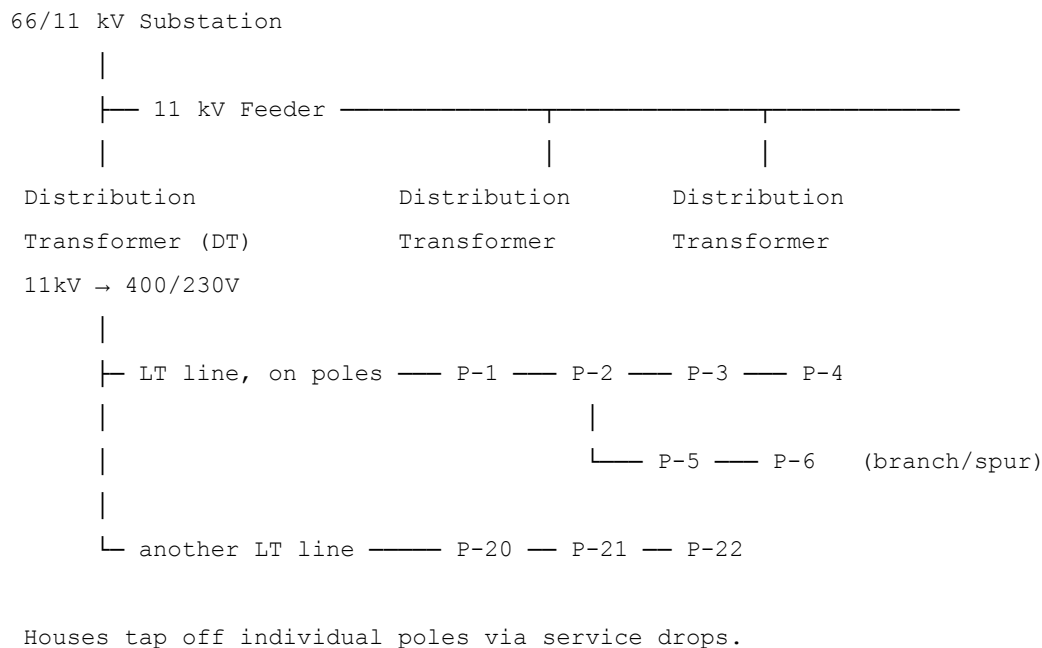
Problem Context — How the Network Works

Read this before you design anything. Most of the difficulty of the assignment comes from the physical shape of the network, and you cannot design the algorithm without understanding it.

Everything below is the domain model for this exercise. It is a simplified but faithful picture of how low-voltage distribution works in Indian cities. Where we have simplified, we say so.

1. The network is a tree, not a mesh

Power reaches a domestic customer through a chain that looks like this:



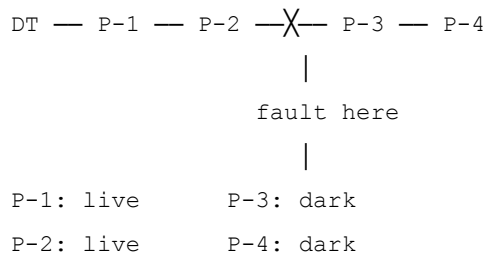
The important property: **the low-tension side is radial**. There are no loops. Every pole has exactly one path back to its distribution transformer, and every DT has exactly one path back to the substation.

That single fact is what makes this problem solvable. It also tells you what shape your core data structure should be.

2. What happens physically when a fault occurs

A fault is almost always on a **span** — the stretch of wire between two adjacent poles — or at a **piece of equipment** (the DT itself, a fuse, a jumper connection at a pole).

When a span fails, everything electrically downstream of it goes dark. Everything upstream stays live.



So the observable signature of a fault is a **boundary**: the last live pole, and the first dark pole beyond it. The fault is on the span between them.

Notice what this means for your algorithm. The fault is on an **edge**. Your sensors report on **nodes**. You are inferring edge state from node state, and the answer is the frontier between the live region and the dark region.

Notice also what it means for alerting. One snapped wire produces dozens of dark poles. All of them are symptoms of a single cause. If your system reports each dark pole as its own incident, you have built a system that makes the control room's night worse.

Multiple simultaneous faults

During a storm, three spans in the same division can fail within minutes. Each one produces its own live/dark boundary. Your localization has to find *all* the boundaries, not just one — and it must not merge two genuinely separate faults into one ticket, or split one fault into two.

Faults that are not on a span

Some things you should expect to see and be able to distinguish:

What broke	What you observe
Span between two poles	Live/dark boundary mid-line
Distribution transformer / its HT fuse	Every pole under that DT goes dark at once, with no live pole beneath it
11 kV feeder	Every pole under every DT on that feeder goes dark
A single pole's own lamp circuit	One pole dark, everything downstream of it still live — not an outage , a broken sensor point

That last row is worth staring at. A single isolated dark pole with live children is physically impossible as a line fault. It is the network telling you your sensor is lying.

3. Why the control room currently takes two hours

The existing process:

1. Householders notice the power is out. Some fraction of them call the complaint number over the next 20–40 minutes.
2. The operator plots the complaints roughly on a paper map or a spreadsheet of ward names and guesses which DT is involved.
3. A lineman is sent to the area on a two-wheeler.
4. He starts at the DT and walks or rides the line, checking poles, until he finds the break. On a long LT line with spurs, this is slow.
5. Only now does the control room know what is actually broken, so only now can they dispatch the right vehicle, ladder, wire, and crew size.

Steps 1–4 are what you are compressing. Step 5 onwards is the department's own operation and is explicitly out of scope.

4. The messy reality you must design around

These are real properties of the deployment, not hypotheticals.

A device that loses power cannot talk for long. Each pole device has a small capacitor-backed reserve. When it loses supply it can transmit for a few seconds — enough for one message. After that it is silent until power returns. Design implications: you may get exactly one "I am dying" packet, or you may get none if the radio was busy. Silence is ambiguous — a pole that stops reporting might be dark, or its modem might be broken.

Some poles have no device at all. Coverage is incomplete. Your topology has gaps, and the fault may be on a span between two poles neither of which reports.

Devices fail on their own. A meaningful fraction of the fleet is offline at any given moment for reasons unrelated to power.

Load shedding is scheduled and routine. Whole feeders are taken down on purpose. These are not faults and must not generate tickets.

Clocks disagree and messages arrive out of order. Two poles that lost power in the same instant may report timestamps a minute apart, and the downstream one may arrive first.

The wiring diagram is incomplete. This is the big one, and it is deliberate. See `02-data-and-systems.md` §3. You know where every pole *is*. You do not reliably know which pole feeds which. For a substantial share of distribution transformers, nobody ever digitized the order of poles along the line.

That last constraint is the heart of the assignment. You cannot walk a tree you do not have. What you do about it — infer it from geography, ask the department for a survey, degrade to a coarser answer, or something we haven't thought of — is the most interesting design decision you will make, and we will read it first.

5. What "good" looks like to the customer

The department will consider this a success if:

- Time from fault occurring to control room knowing the location is **under two minutes**, versus two hours today.
- The location is precise enough to drive to and hand to a crew — a span, not a ward.
- Operators trust the alerts. A system that fires on load shedding and dead modems gets ignored, and an ignored system has zero value.
- Ticket closure reflects reality. "Fixed" means the poles are live again, as measured, not as claimed.

Now go read `02-data-and-systems.md` (`02-data-and-systems.md`) for the actual data contracts.